SUPPORTING ADULT GUIDANCE

Negative and fractional indices

Number Systems and Calculator Judgement · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson extends powers to zero, negative and fractional indices. Learners first see the index laws produce 5^{-2} while cancelling gives 1/25. On a power ladder they step down past power 1, finding $a^0 = 1$ and negative powers as reciprocals, and then step sideways onto half and third rungs to find roots. Each part has a stretch proof, using the index laws, so the meanings are justified rather than just pattern-spotted.

Driving Question: *What can a power of –2 or ½ possibly mean?*

The mathematics, in plain terms

Positive whole-number powers are repeated multiplication. Zero, negative and fractional indices are **defined** so that the index laws (add when multiplying, subtract when dividing, multiply for a power of a power) stay true. From $a^m \div a^m = 1$ we get $a^0 = 1$ (so $a \neq 0$); from $a^0 \div a^n$ we get $a^{-n} = 1/a^n$; from $(a^{1/n})^n = a$ we get $a^{1/n}$ = the nth root of a. For fractional indices the base is kept positive, and $a^{½}$ means the positive root.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Down the ladder, and why

Work out 2^0 and 2^{-3}, then prove that $a^0 = 1$.

1. $2^2 = 4$, $2^1 = 2$. Down one rung: ÷ 2.	$2^0 = 2 \div 2 = 1$
2. Keep dividing by 2.	$2^{-1} = 1/2$, $2^{-2} = 1/4$
3. One more rung.	$2^{-3} = 1/8 = 1/2^3$
4. Proof: by the law, $a^3 \div a^3 = a^0$. But a number divided by itself is 1.	$a^0 = 1$ ($a \neq 0$)

$2^0 = 1$ and $2^{-3} = 1/8$: the reciprocal of 2^3.

Example 2 — Sideways on the ladder

Work out $8^{2/3}$ and $8^{-2/3}$.

1. Three third-steps make one whole step, × 8.	? × ? × ? = 8
2. 2 × 2 × 2 = 8, so each third-step is × 2.	$8^{1/3} = \sqrt[3]{8} = 2$
3. Two third-steps up from 1: × 2, × 2.	$8^{2/3} = (\sqrt[3]{8})^2 = 4$
4. A negative index: the reciprocal.	$8^{-2/3} = 1/4$

$8^{2/3} = 4$ and $8^{-2/3} = 1/4$. The root first, then the power, then (if negative) the reciprocal.

Pattern, then proof

The ladder makes each meaning feel natural: dividing by the base on every step down. But a pattern only suggests. The 'Why must it be?' stretch in each ladder part proves that no other value keeps the index laws true, and shows the scope (why $a \neq 0$; why the root is positive). Encourage learners who are ready to try the proofs; for others, the ladder and the reciprocal idea are the secure core. Calculators are not used: exact values are the point.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Two answers to one question: the index laws give 5^{-2}, cancelling gives 1/25.	~6 min
Movement	Down the power ladder: power 0 and negative powers. Stretch: prove $a^0 = 1$.	~12 min
Reflection	Reciprocal, not negative: exact values with misconception-aware feedback.	~8 min
Creativity	Sideways on the power ladder: half and third rungs, build your own. Stretch: prove $a^{½} = \sqrt{a}$.	~14 min
Rest	The ladder running down for ever, never reaching 0.	~2 min
Nutrition	Intellectual: the satisfaction of a definition that is forced.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking a negative index makes a negative answer ($5^{-2} = -25$).
Ask: “Go down the ladder from $5^0 = 1$. Does dividing ever go below 0?”

Multiplying the base by the index ($5^{-2} = -10$).
Ask: “What does 5^2 mean? So what might 5^{-2} be?”

Believing $a^0 = 0$.
Ask: “What is one rung below a^1?”

Halving the base for a power of ½ ($9^{½} = 4.5$).
Ask: “What is 4.5 × 4.5? Is that 9?”

Doing the power before the root for $a^{m/n}$, making the numbers huge.
Ask: “Which is easier to find first?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the index laws for whole-number powers, and squares, cubes and their roots. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says $9^{½} = 4.5$ because “half of 9 is 4.5”. The aim is a question that lets them find the slip themselves.

- “On the ladder, what does one whole step from 9^0 to 9^1 multiply by?”

- “If two equal half-steps make that, what does each half-step multiply by?”
- “Check: what is 4.5 × 4.5?”
- “So what must $9^{½}$ be?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Are the proofs required?
They are a stretch. The secure core is the ladder and the reciprocal idea; the proofs deepen understanding for learners who are ready.

Why is $a^{½}$ the positive root?
Both 3 and –3 square to 9, but $a^{½}$ is defined as the positive root, matching √. This keeps it a single value.

What about 0^0?
The proof needs $a \neq 0$, because 0 ÷ 0 has no value. 0^0 is left undefined at this stage.

What comes next?
Lesson 5: surds — meaning and simplifying.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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